An Intro to Natural Language Processing

GENERAL ASSEMBLY

Agenda

- What is NLP and what is it good for?
- Some NLP vocabulary
- Text Preprocessing
- Sentiment analysis

What is Natural Language?

"To understand text analytics and natural language processing, we need to understand what makes a language "natural." In simple terms, ***natural language*** *is one developed and evolved by humans through natural use and communication, rather than constructed and created artificially, like a computer programming language."*

- Dipanjan Sarkar, *Text Analytics with Python: A Practitioner's Guide to NLP*

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What is Natural Language Processing?

Natural Language Processing (NLP) is a specialized field of computer science primarily concerned with designing and building applications and systems for facilitating the interaction between machines and the natural language of humans.

[Definition paraphrased from Sarkar.]

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What is NLP used for?

NLP has a several applications, including:

- **Machine translation**: Providing a systematic, grammatically correct translation between two natural languages
- **Speech recognition**: *“Siri, play Despacito.”*
- **Question and Answering Systems**: *“Siri, what is the capital of France?”*
- **Text summarization**: Take a piece of text and condense it to a much smaller version that still contains the basic information
    - Sentiment analysis is a type of this!
- **Text categorization**: Given a document, does it belong to category A or B?
    - e.g. Is this email *spam* or *ham*?
    - You’ll do this in the next lesson!
- **Text clustering**: Grouping documents based on their overall vibe
    - Outside the scope of our class… but only barely!

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Some NLP Vocab

NLP Vocab I

- A **document** is a piece of text that represents one item of analysis
  - For example, this could be a Yelp review, a Tweet, or a news article

- A group of documents is called a **corpus**. You can think of this as like a dataset of documents. The plural of corpus is **corpora**.

- A document is further broken down into **tokens**, the basic unit of analysis we're interested in. Generally, tokens are just **words**. The act of converting a document into tokens is **tokenizing**.

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Tokens

For many NLP tasks, **tokens** are just **words**.

Tokens can be smaller than words, such as **characters** or **syllables** for some tasks, such as text-to-speech.

In our class, we'll also see ***n*-grams** as tokens

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n-grams

An ***n*-gram** is a tuple of *n* words used as a whole token. For example:

original document	the cat sat on the mat
unigrams (words)	the, cat, sat, on, the, mat
bigrams	(the, cat), (cat, sat), (sat, on), (on, the), (the, mat)
trigrams	(the, cat, sat), (cat, sat, on), (sat, on, the), (on, the, mat)

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NLP Vocab II

Traditional, tabular data:

*“We’re going to **preprocess** our **dataset**.”*

Text data:

*“We’re going to do **text normalization** on our **corpus**.”*

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What do you think are examples of text normalization? (preprocessing steps applied to natural language data)

Text Normalization

The **normalization** (preprocessing) of your corpus could involve the following steps:

- Tokenization
- Removing or replacing special characters and punctuation ($#@%&, etc)
- Expanding contractions (don't ↦ do not)
- Removing **stop words**
- Correcting repeating characters/typos (heyyy ↦ hey)
- **Stemming** our tokens
- **Lemmatizing** our tokens
- **Part of Speech Tagging**

Stop Words

In English, **stop words** are small words that have little or no significance to the meaning of the overall sentence.

Original Sentence	*I am eating a sandwich from the restaurant by the store*
After stop word removal	*eating sandwich restaurant store*

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Stemming

To **stem** a word, we remove its suffixes so that multiple forms of the words are recognized by their root.

- jumps ↦ jump
- jumped ↦ jump
- jumping ↦ jump

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Stemming

Here's our example using the most popular **Porter stemmer**. Note that the stemmer is algorithmic and not perfect.

Original Sentence	*I am eating a sandwich from the restaurant by the store*
After stop word removal	*eating sandwich restaurant store*
After stemming	*eat sandwich restaur store*

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Lemmatizing

Similar to stemming, **lemmatizing** reduces a word to its base form. It's a lot smarter in that it can conjugate verbs and has more rules, but is therefore much slower.

- going ↦ go
- went ↦ go
- gone ↦ go

Lemmatizing

Another major issue with lemmatizing is that you need to know a word's **part of speech** or else the algorithm won't know how to conjugate

- wound ↦ wind (as a verb)
- wound ↦ wound (as a noun)

Lemmatizing

Here's our example again using the popular **WordNet lemmatizer**.

Original Sentence	*I am eating a sandwich from the restaurant by the store*
After stop word removal	*eating sandwich restaurant store*
After lemmatizing	*eat sandwich restaurant store*

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Aside: Bag-of-words

It turns out, the order of words in a sentence (as dictated by English grammar) isn't hugely impactful to our understanding of a sentence!

Many NLP models are **bag-of-words models**. That is, just like words in a bag, the models won't take word order into account!

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Bag-of-words examples

What is the overall sentiment of this *unordered, normalized* sentence?

french omelette restaurant disgusting

Bag-of-words examples

What is the overall sentiment of this *unordered, normalized* sentence?

french omelette restaurant disgusting

It was actually:

The omelette at this French restaurant was disgusting!

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Bag-of-words examples

How about another:

quiche spinach not try bad

Bag-of-words examples

How about another:

quiche spinach not try bad

It was actually:

Try the spinach quiche, it was not bad!

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Bag-of-words examples

How about another:

quiche spinach not try bad

It was actually:

Try the spinach quiche, it was not bad!

but it also could have been...

Do not try the spinach quiche, it was bad!

Sentiment Analysis

Sentiment Analysis

A common and simple NLP task is **sentiment analysis**. That is, assigning a sentiment value to a given document based on how positive or negative it is. For example:

Negative sentiment document: *This was a terrible movie, it was so bad I wanted to vomit!*

Positive sentiment document: *Our server was very courteous, and the meal was delicious!*

Neutral sentiment document: *I have tickets for the 2pm train.*

Sentiment Analyzers

Most sentiment analysis models are bag-of-words models that assign point values to individual words. Here are those examples passed through the VADER lexicon sentiment analyzer. VADER was fit to work best with social media text.

Document	VADER Lexicon Sentiment
This was a terrible movie, it was so bad I wanted to vomit!	`{'neg': 0.458, 'neu': 0.542, 'pos': 0.0, 'compound': -0.822}`
Our server was very courteous, and the meal was delicious!	`{'neg': 0.0, 'neu': 0.513, 'pos': 0.487, 'compound': 0.8217}`
I have tickets for the 2pm train.	`{'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}`

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Python Time!



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